# Aritha Munasinghe

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## Education

**California State University, Sacramento** — Sacramento, CA

*Bachelor of Science in Computer Science* — *Anticipated Graduation: December 2027*

- **GPA:** 4.0/4.0 (Dean's Honor List) | **Coursework:** Data Structures & Algorithms, Computer Software Engineering, System Programming (Unix), Computer Architecture, Applied Linear Algebra

## Experience

**Sacramento Municipal Utility District (SMUD)** — Sacramento, CA

*STEM Student Assistant – Enterprise Applications Engineering* — *May 2026 – Present*

- Own production monitoring for **Kubernetes** services across 11 workload groups and 21 pods, scoping requirements with the engineers who use it and carrying it through deployment and ongoing operation.
- Shipped a second version after the first produced false positives during rolling deployments, replacing name-prefix matching with a controller-aware grouped **KQL** query that extended one rule to every monitored workload group.
- Automated **Windows** patching for an Esri ArcGIS Server deployment with a script that switches primary and standby nodes through failover, so each server can be updated in turn.
- Root-caused a stalled **Kubernetes** rollout as a persistent-volume attachment conflict between terminating and incoming pods, clearing a deployment that had resisted repeated Helm release attempts.

**DRM Electronics** — Colombo, Sri Lanka

*Junior Software Engineer* (promoted after 4 months) — *Dec. 2023 – Jul. 2024*

- Refactored legacy **Node.js** backend APIs to raise sustained concurrent request throughput across a multi-tier service architecture under live traffic.
- Optimized **PostgreSQL** schemas and rewrote reporting queries against high-load analytical modules, cutting retrieval latency on the heaviest read paths.

*Software Engineering Intern* — *Aug. 2023 – Dec. 2023*

- Ramped onto an existing production codebase and shipped front-end improvements in **Angular**, restructuring component rendering on the views users hit most.

## Projects

**Real-Time Collaborative Editor** | *React, TypeScript, Python, FastAPI, Docker* — Aug. 2025 – Present

- Built a multi-user editor with a **React**/TypeScript client over a **Python**/FastAPI backend, keeping concurrent users consistent over Socket.IO by eliminating an event echo loop and duplicate listener registration.
- Closed a path where an unauthenticated client could join any room by id and read its live contents, refactoring token validation into a standalone verifier and covering it with a 12-test **pytest** suite.
- Replaced a third-party code-execution API with an in-house **Docker** sandbox (no network, memory/CPU/PID limits, host-enforced timeouts), capping concurrency with a semaphore that queues rather than rejects, validated by an 11-attack adversarial test suite.

**Generative AI Knowledge Retrieval Engine** | *Python, FastAPI, PostgreSQL (pgvector)* — Dec. 2025

- Built a document ingestion and retrieval service in **Python**/FastAPI that chunks unstructured files into **PostgreSQL** with pgvector for embedding storage and nearest-neighbour search.
- Grounded generated answers in retrieved passages rather than model recall alone, so every response traces back to a source document in the corpus.

## Technical Skills

**Languages**: Python, Java, JavaScript (ES6+), TypeScript, SQL, C, Bash, PowerShell
**Back End & Data**: FastAPI, Node.js, REST APIs, WebSockets, PostgreSQL, query optimization, pgvector
**Infrastructure**: Docker, Kubernetes, Azure, Linux, Windows Server, CI/CD, GitHub Actions, Git, monitoring
**Testing & Quality**: pytest, mypy, unit and regression testing, adversarial testing, code review
**Front End & AI Tools**: React, Angular, Claude Code, Copilot | **Certification**: Harvard CS50x